

Unit 1 Test

Dynamics

K	T	A	C
16/18	17 ^{1/4} /18	9 ^{1/2} /10	9 ^{1/4} /10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

THINKING

1. The blades of a blender of radius 50 mm are spinning at a constant speed of 1.2×10^3 cm/s. What is the centripetal acceleration of a single point on the edge of one of the blades?
Show all your work.

$$r = 50 \text{ mm} = 5.0 \text{ cm} = 0.05 \text{ m}$$

$$v = 1.2 \times 10^3 \text{ cm/s} = 1200 \text{ cm/s} = 12 \text{ m/s}$$

$$a_c = \frac{v^2}{r}$$

$$a_c = \frac{12^2}{0.05}$$

$$a_c = 2880 \text{ m/s}^2$$

missing
direction



Final answer: $a_c = 2880 \text{ m/s}^2$

[1.5 marks]

2. A speed boat travelling in a uniform circular motion. The boat is undergoing centripetal acceleration of 13.7 m/s^2 . The radius of the boat's path is 0.122 km . Calculate the boat's speed in km/h.
Show all your work.

$$a_c = 13.7 \text{ m/s}^2$$

$$r = 0.122 \text{ km} = 122 \text{ m}$$

$$v = ?$$

$$a_c = \frac{v^2}{r}$$

$$v^2 = a_c r$$

$$v = \sqrt{a_c r}$$

$$v = \sqrt{13.7 \cdot 122}$$

$$v = 40.88 \text{ m/s} \times \frac{3.6 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}}$$

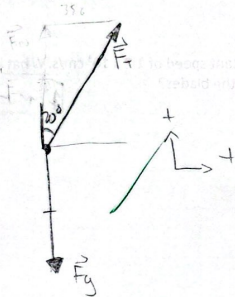
$$v = 147.18 \text{ km/h}$$

Final answer: $v = 147.18 \text{ km/h}$

3. You are on a subway holding yourself off the floor by pulling on an overhead rail. The subway starts moving, and your body makes an angle of 20.0° with the vertical.

- a) Draw a well labelled free body diagram (FBD) for this situation.
 b) Determine the acceleration of the subway car.
 Show all your work.

[4 marks]



$$\vec{F}_{\text{net},y} = 0$$

$$\vec{F}_{\text{net},y} = \vec{F}_T - \vec{F}_g$$

$$\vec{F}_T = \vec{F}_{\text{net},y} + \vec{F}_g$$

$$\vec{F}_T = -\vec{F}_g$$

$$\vec{F}_g = -mg$$

$$\vec{F}_T = -mg$$

$$\vec{F}_{Tx} = \vec{F}_T \tan(20^\circ)$$

$$\vec{F}_{Tx} = -mg \tan(20^\circ)$$

$$\vec{F} = m\vec{a}$$

$$\vec{a}_x = \frac{\vec{F}_{Tx}}{m}$$

$$\vec{a}_x = \frac{-mg \tan(20^\circ)}{m}$$

$$\vec{a}_x = -g \tan(20^\circ)$$



$$\vec{a}_x = (-9.8 \text{ m/s}^2) \tan(20^\circ)$$

$$\vec{a}_x = 3.567 \text{ m/s}^2 [\text{F}]$$

Final answer: $\vec{a}_x = 3.567 \text{ m/s}^2 [\text{F}]$

Name: _____

Date: _____

4. A small remote-controlled car with mass 1.60 kg moves at a constant speed of 12.0 m/s in a track formed by a vertical circle inside a hollow metal cylinder that has a radius of 5.00 m as shown in the diagram. Determine the magnitude of the normal force exerted on the car by the walls of the cylinder at:

a) Point A (bottom of the track)

b) Point B (top of the track).

Show all your work.

$m = 1.60 \text{ kg}$
 $v = 12.0 \text{ m/s}$
 $r = 5.00 \text{ m}$

a)

$$F_c = \frac{mv^2}{r}$$

$$F_c = F_N - F_g$$

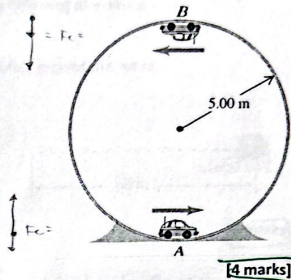
$$F_N = F_c + F_g$$

$$F_g = mg$$

$$F_N = \frac{mv^2}{r} + mg$$

$$F_N = \frac{1.6 \cdot 12^2}{5} + 1.6 \cdot 9.81$$

$$F_N = 61.776 \text{ N [u]}$$



[4 marks]

b)

$$F_c = \frac{mv^2}{r}$$

$$F_c = F_g + F_N$$

$$F_N = F_c - F_g$$

$$F_g = mg$$

$$F_N = \frac{mv^2}{r} - mg$$

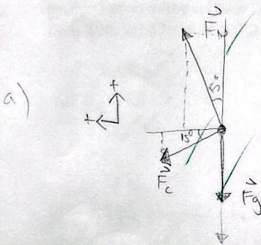
$$F_N = \frac{1.6 \cdot 12.0^2}{5} - 1.6 \cdot 9.81$$

$$F_N = 30.384 \text{ N [D]}$$

Final answer: a) 61.776 NFinal answer: b) 30.384 N

5. A car travels on dry, banked highway ramp of radius 85 m is experiencing friction. The coefficient of static friction between the tires and the road is 0.70. The banking angle is 15° to the horizontal.

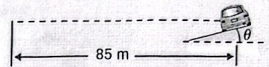
- a) Draw a well labelled free-body diagram (FBD) of the car on the banked curve.
 b) Determine the equation of the net force acting on the car in the following directions:
 i) the vertical direction.
 ii) the horizontal direction.
 c) Determine the constant speed the car must maintain to travel safely around the curve.
 Show all your work.



$$\mu_s = 0.7$$

$$r = 85 \text{ m}$$

$$\theta = 15^\circ$$



[6 marks]

b) $F_{\text{net}y} = 0 \text{ N}$

$$F_{\text{net}y} = F_{\text{N}y} - F_{\text{G}}$$

$$0 = F_{\text{N}y} - F_{\text{G}}$$

$$F_{\text{N}y} = F_{\text{G}}$$

$$F_{\text{N}y} = F_{\text{N}} \cos 15^\circ$$

$$F_{\text{G}} = F_{\text{N}} \sin 15^\circ = \mu_s F_{\text{N}} \sin 15^\circ$$

$$F_{\text{N}} = \frac{mg}{\sin 15^\circ}$$

i) $F_{\text{net}x} = F_{\text{N}} \sin 15^\circ - \mu_s F_{\text{N}} \cos 15^\circ - mg$

$$F_{\text{net}x} = F_{\text{N}x} - F_{\text{G}x}$$

$$F_{\text{N}x} = F_{\text{N}} \sin 15^\circ$$

$$F_{\text{G}x} = F_{\text{G}} \cos 15^\circ = \mu_s F_{\text{N}} \cos 15^\circ$$

ii) $F_{\text{net}x} = F_{\text{N}} \sin 15^\circ + \mu_s F_{\text{N}} \cos 15^\circ$

c) $0 = F_{\text{N}} \cos 15^\circ + \mu_s F_{\text{N}} \sin 15^\circ - mg$

$$mg = F_{\text{N}} (\cos 15^\circ + \mu_s \sin 15^\circ)$$

$$F_{\text{N}} = \frac{mg}{(\cos 15^\circ + \mu_s \sin 15^\circ)}$$

$$F_{\text{net}x} = \frac{mv^2}{r}$$

$$\frac{mv^2}{r} = F_{\text{N}} \sin 15^\circ + \mu_s F_{\text{N}} \cos 15^\circ$$

$$\frac{mv^2}{r} = F_{\text{N}} (\sin 15^\circ + \mu_s \cos 15^\circ)$$

$$v = \sqrt{\frac{r F_{\text{N}} (\sin 15^\circ + \mu_s \cos 15^\circ)}{m}}$$

$$v = \sqrt{\frac{r mg}{(\cos 15^\circ + \mu_s \sin 15^\circ)} (\sin 15^\circ + \mu_s \cos 15^\circ)}$$

$$v = \sqrt{\frac{gr (\sin 15^\circ + \mu_s \cos 15^\circ)}{(\cos 15^\circ + \mu_s \sin 15^\circ)}}$$

$$v = \sqrt{\frac{9.81 \cdot 85 (\sin 15^\circ + 0.7 \cos 15^\circ)}{(\cos 15^\circ + 0.7 \sin 15^\circ)}} \quad v = 31.52 \text{ m/s}$$

Final answer: b i) _____

Final answer: b ii) _____

Final answer: c) $v = 31.52 \text{ m/s}$

Unit 1 Test

Dynamics

K	T	A	C
/ 18	/ 18	9 ^{1/2} / 10	9 ^{1/4} / 10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. The drawing shows a baggage carousel at an airport. Your suitcase has not slid all the way down the slope and is going around at a constant speed on a circle ($r = 11.0 \text{ m}$) as the carousel turns. The coefficient of static friction between the suitcase and the carousel is 0.760, and the angle θ in the drawing is 36.0° . How much time is required for your suitcase to go around once? *Show all your work.*

$r = 11 \text{ m}$

$\mu_s = 0.76$

$\theta = 36^\circ$

$\vec{F}_{\text{net}y} = 0 \text{ N}$

$\vec{F}_{\text{net}y} = \vec{F}_s \sin 36^\circ + \vec{F}_N \cos 36^\circ - \vec{F}_g$

$\vec{F}_g = m\vec{g}$

$0 \text{ N} = \vec{F}_s \sin 36^\circ + \vec{F}_N \cos 36^\circ - m\vec{g}$

$\vec{F}_s = \vec{F}_N \mu_s$

$m\vec{g} = \vec{F}_N \mu_s \sin 36^\circ + \vec{F}_N \cos 36^\circ$

$m\vec{g} = \vec{F}_N (\mu_s \sin 36^\circ + \cos 36^\circ)$

$\vec{F}_N = \frac{m\vec{g}}{(\mu_s \sin 36^\circ + \cos 36^\circ)}$

$\vec{F}_{\text{net}x} = \vec{F}_s \cos 36^\circ - \vec{F}_N \sin 36^\circ$

$\vec{F}_{\text{net}x} = \frac{mv^2}{r}$

$\vec{F}_s = \mu_s \vec{F}_N$

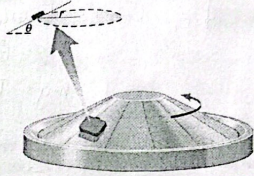
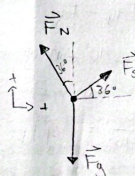
$\frac{mv^2}{r} = \mu_s \vec{F}_N \cos 36^\circ - \vec{F}_N \sin 36^\circ$

$\frac{mv^2}{r} = \vec{F}_N (\mu_s \cos 36^\circ - \sin 36^\circ)$

$v = \sqrt{\left(\frac{m\vec{g}}{(\mu_s \sin 36^\circ + \cos 36^\circ)} \right) \left(\frac{r (\mu_s \cos 36^\circ - \sin 36^\circ)}{m} \right)}$

$v = \sqrt{\left(\frac{9.8 \text{ m/s}^2}{(0.76 \sin 36^\circ + \cos 36^\circ)} \right) (11.0 \text{ m} (0.76 \cos 36^\circ - \sin 36^\circ))}$

$v = 1.5241 \text{ m/s}$



[5 marks]

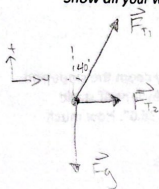
$T = \frac{2\pi r}{v}$

$T = \frac{2 \cdot \pi \cdot 11}{1.5241}$

$T = 45.34 \text{ s}$

Final answer: $T = 45.34 \text{ s}$

2. In a Giant Swing at a county fair, as shown in the diagram, a seat is connected to two cables, one of which is horizontal. The seat swings in a horizontal circle at a rate of 28.0 rpm (rev/min). If the combined mass of the person and the seat is 110.2 kg, determine the tension in each cable, the horizontal cable and the angled cable. Show all your work.



$$f = 28 \text{ rpm} = 0.4667 \quad r = 7.5 \text{ m}$$

$$m = 110.2 \text{ kg}$$

$$\vec{F}_{\text{net},y} = 0$$

$$\vec{F}_{\text{net},y} = F_{T1y} - F_g$$

$$0 = F_{T1} \cos 40^\circ - mg$$

$$mg = F_{T1} \cos 40^\circ$$

$$F_{T1} = \frac{mg}{\cos 40^\circ} = 1411.23 \text{ N}$$

$$\vec{F}_{\text{net},x} = \vec{F}_{T1x} + \vec{F}_{T2}$$

$$F_{\text{net},x} = m(4\pi^2 f^2 r)$$

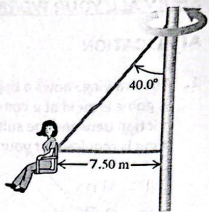
$$m(4\pi^2 f^2 r) = F_{T1} \sin 40^\circ + F_{T2}$$

$$F_{T2} = m(4\pi^2 f^2 r) - \left(\frac{mg}{\cos 40^\circ} \right) \sin 40^\circ$$

$$F_{T2} = 110.2(4\pi^2 \cdot 0.4667^2 \cdot 7.5) - \left(\frac{110.2 \cdot 9.81}{\cos 40^\circ} \right) \sin 40^\circ$$

$$F_{T2} = 6199.75 \text{ N}$$

$$F_{T1}$$



[5 marks]

Wic meeres!

$$\vec{F}_{T2}$$

Final answer: 6199.75 N CF

Final answer: 1411.23 N CU 44F3

$$\vec{F}_{T1}$$

missing & i.e. hand.



COMMUNICATION

1. Will the acceleration of a car be the same when a car travels around a sharp circular curve at a constant 60 km/h as when it travels around a gentle (wider) curve at the same speed? Explain.

Centripetal acceleration is based on the formula $a_c = \frac{v^2}{r}$, and a sharper curve means the effective radius is smaller. a_c is inversely proportional to radius, meaning a smaller radius will increase the acceleration of the car. The gentler curve will have a lower acceleration. (2 marks)

2. If there is a net force on an object in uniform circular motion, why doesn't the object's speed change?

Uniform circular motion has an object which is experiencing a net force towards the center of rotation. The velocity is tangent to the circular path, and the acceleration is pointed to the middle, meaning the speed won't change but the velocity and direction changes. (2 marks)

- 3a. Two identical cars, one on the moon and one on the earth, have the same speed and are rounding banked turns that have the same radius r . There are two forces acting on each car, its weight mg and the normal force F_N exerted by the road. Consider that the weight of an object on the moon is about one-sixth of its weight on earth. How does the centripetal force on the moon compare with that on the earth? Explain. (1 mark)

Centripetal force can be calculated by combining the horizontal component of F_N and the gravitational force downward. Because the equation for centripetal force requires gravity, it would be one-sixth. (2 marks)

- 3b. Other things being equal, would it be easier to drive at high speed around an unbanked horizontal curve on the moon than to drive around the same curve on the earth? Explain. (2 marks)

The only force for centripetal force would be the F_N from the tires. This is derived from $F_N = \frac{mv^2}{r}$ and $F_N = F_g$. This means as gravity decreases, F_N would decrease. Because the moon's gravity is $\frac{1}{6}$ th of Earth's, it's F_g would be lower, thus meaning the F_N would be lower, thus finally lowering F_N , making it harder to corner due to lower centripetal forces on the moon. (2 marks)

	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/physical conventions are used correctly. [1/2]	Some mathematical/physical conventions are used correctly. [1]	Most mathematical/physical conventions are used correctly. [1-1/2]	Consistently uses mathematical/physical conventions correctly. [2]